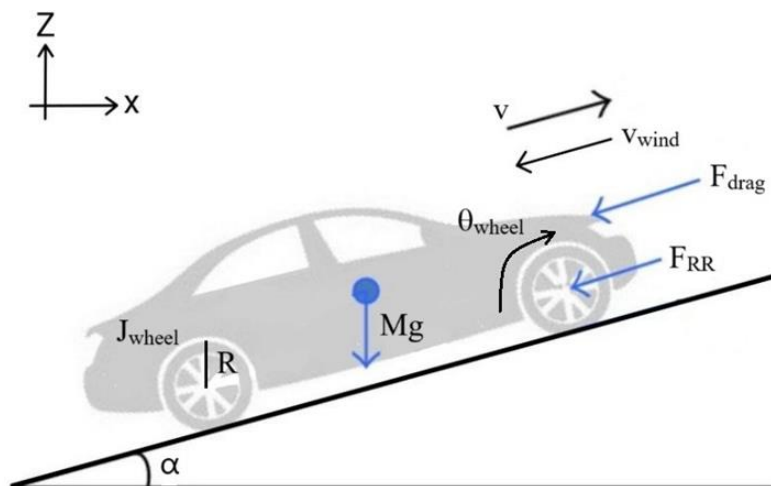




POLITECNICO
MILANO 1863

CONTROL OF MECHANICAL SYSTEMS

Cruise control project



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Part 1:

Assumptions:

- Non-deformable shaft ($k_t = \infty$)
- 1 dof system (we select the speed of the vehicle v)
- Small oscillations around the steady state condition: $v_0 = 15 \frac{m}{s}$ and $v_{wind,0} = 3.5 \frac{m}{s}$

Lagrange approach:

$$E_k = \frac{1}{2} J_m \omega_M^2 + \frac{1}{2} J_\omega \omega_\omega^2 \cdot 4 + \frac{1}{2} M v^2 \Rightarrow E_k = \frac{1}{2} \left(\frac{J_M \tau_1^2}{R^2} + \frac{4 J_\omega}{R^2} + M \right) v^2 = \frac{1}{2} m^* v^2$$

$$D = \frac{1}{2} (r_1 + r_2) \omega_T^2 \Rightarrow D = \frac{1}{2} \left(\frac{r_1 + r_2}{R^2} \right) v^2 = \frac{1}{2} r^* v^2$$

$$V = 0$$

$$\delta W = \vec{T}_M \cdot \partial \vec{\theta}_M + \vec{F}_{Drag} \cdot \partial \vec{x} + \vec{F}_g \cdot \delta \vec{h}_g + \vec{F}_{Roll} \cdot \partial \vec{x} \Rightarrow \left(T_M \frac{\tau_1}{R} - F_{Drag} - Mg \sin a - F_{Roll} \right) \delta x \Rightarrow Q \delta x$$

(As $\tau_2 = 1$ we neglect it in the formulas directly)

We get the expression for the non-linear equation of motion of the system:

$$m^* \dot{v} + r^* v = T_M \frac{\tau_1}{R} - F_{Drag} - Mg \sin a - F_{Roll}$$

By considering $T_M = T_{M_0} + T_C$:

- T_{M_0} represent the engine's torque provided in such a way as to allow the vehicle to maintain a speed equal to v_0
- T_C represent the control torque, i.e. the torque that the engine provides to face external disturbances and keep the vehicle's speed at its steady state value

T_{M_0} value at steady state condition:

$$\begin{cases} v_{wind} = v_{wind_0} \\ v = v_0 \\ \dot{v} = 0 \\ T_C = 0 \end{cases} \Rightarrow r^* v_0 = T_{M_0} \frac{\tau_1}{R} - \frac{1}{2} \rho_A A_f C_x (v_0 + v_{wind_0})^2 - Mg \sin a - C_{RR} Mg \cos a (1 + k_{RR} v_0)$$

$$T_{M_0} = 183.06 \text{ Nm}$$

Linearization in the neighbourhood of the steady state condition $v = v_0$, $v_{wind} = v_{wind_0}$:

$$v = \tilde{v} + v_0 \quad \tilde{\dot{v}} = \dot{v}$$

$$v_{wind} = v_{wind_0} - \tilde{v}_{wind} \quad (v_{wind} < 0)$$

$$F_{Drag} \cong F_{D_0} + \left. \frac{\delta F_D}{\delta v} \right|_{v_0, v_{wind_0}} (v - v_0) + \left. \frac{\delta F_D}{\delta v_{wind}} \right|_{v_0, v_{wind_0}} (v_{wind_0} - v_{wind})$$

$$-K_1 = \left. \frac{\delta F_D}{\delta v} \right|_{v_0, v_{wind_0}} \quad \text{and} \quad -K_2 = \left. \frac{\delta F_D}{\delta v_{wind}} \right|_{v_0, v_{wind_0}} \Rightarrow = F_{D_0} - K_1 \tilde{v} - K_2 \tilde{v}_{wind}$$

F_{Roll} is already a linear function.

Linearized equation of motion of the system: $m_{TOT}\ddot{v} + r_{TOT}\dot{v} = T_C \frac{\tau_1}{R} + K_2 \tilde{v}_{wind}$

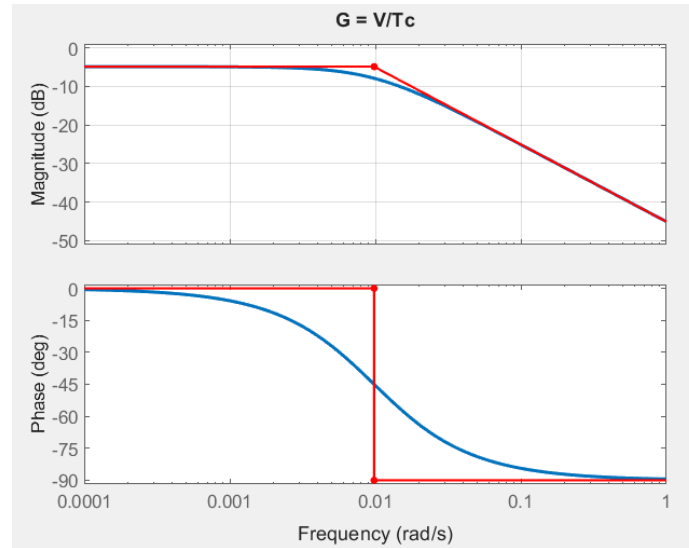
$$\text{where: } \begin{cases} m_{TOT} = m^* \\ r_{TOT} = r^* - K_1 + C_{RR} Mg \cos \alpha k_{RR} \end{cases}$$

Transfer functions:

$$D(s) = \frac{V}{V_{wind}} = \frac{+K_2}{(m_{TOT}s + r_{TOT})}$$

$$G(s) = \frac{V}{T_C} = \frac{\tau_1}{R} \cdot \frac{1}{(m_{TOT}s + r_{TOT})}$$

Poles of G(s): $p = -\frac{r_{TOT}}{m_{TOT}} = -0.0098$, $\text{Re}(p) < 0$: the uncontrolled system is stable



Part 2:

$T_C(t)$ is now controlled by a PI regulator: $T_C = k_p(v_{Ref} - \tilde{v}) + k_i \left(\int_0^{+\infty} (v_{Ref} - \tilde{v}) dt \right)$

Cruise control is based on the vehicle's ability to adapt the control torque T_C , delivered by the engine, to possible external disturbances, which can influence the speed of the vehicle, and change it moment by moment to ensure that the desired speed is maintained.

We start from a condition where the vehicle's speed is settled at its steady state value v_0 ; the cruise control must work in order to keep it unchanged $\rightarrow v_{Ref} = 0$

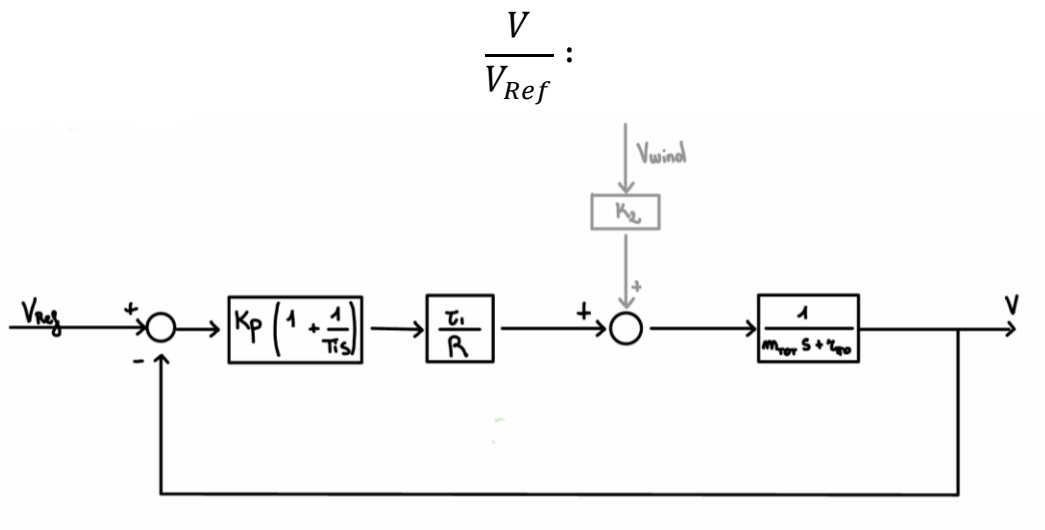
Laplace domain:

$$T_C = k_p(V_{Ref} - V) + \frac{k_i}{s}(V_{Ref} - V), \quad k_i = \frac{k_p}{T_i} \quad \text{and} \quad V_{Ref} = 0 \Rightarrow T_C = \left(k_p + \frac{k_i}{s}\right)(-V)$$

$$\begin{cases} (m_T s + r_T)V = T_c \frac{\tau_1}{R} + K_2 V_{wind} \\ T_c = \left(k_p + \frac{k_i}{s}\right)(-V) = -k_p \left(\frac{1 + T_i s}{T_i s}\right)V \end{cases}$$

Stability analysis:

We define: $k_p \left(\frac{1+T_i s}{T_i s}\right) = R$, $\frac{\tau_1}{R} = K$, $\frac{1}{(m_{TOT} s + r_{TOT})} = \tilde{G}$



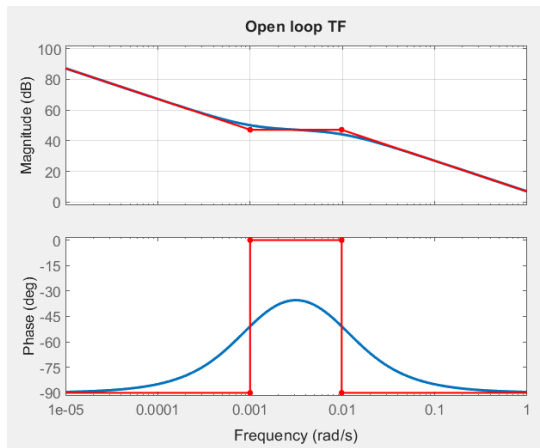
- *Open loop:* $KR\tilde{G} = \frac{\tau_1}{R} \cdot k_p \left(\frac{1+T_i s}{T_i s}\right) \cdot \frac{1}{(m_{TOT} s + r_{TOT})}$
 1 zero: $z = -\frac{1}{T_i}$ (stable)
 2 poles: $p_1 = 0$, $p_2 = -\frac{r_{TOT}}{m_{TOT}}$ (stable)
- *Closed loop:* $L(s) = \frac{V}{V_{Ref}} = \frac{KR\tilde{G}}{1+KR\tilde{G}} = \frac{\tau_1 k_p (1+T_i s)}{s^2(m_{TOT} T_i R) + s(r_{TOT} T_i R + \tau_1 k_p T_i) + \tau_1 k_p}$
 1 zero and 2 poles

We analyse the stability of the system by applying some undirect and direct methods (margin, Nyquist and root locus)

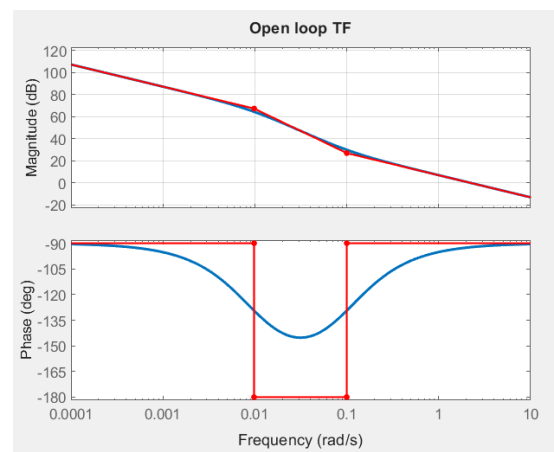
There may be two different situations:

1. $z < p_2$ ($T_i > 102$)
2. $z > p_2$ ($T_i < 102$)

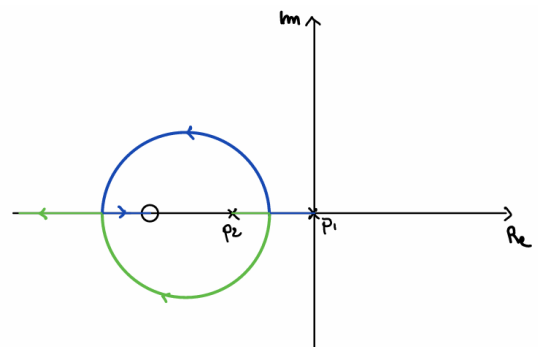
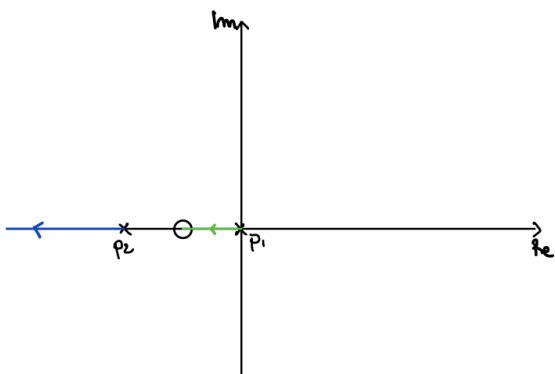
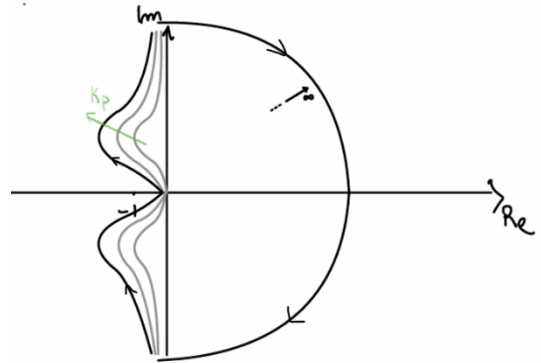
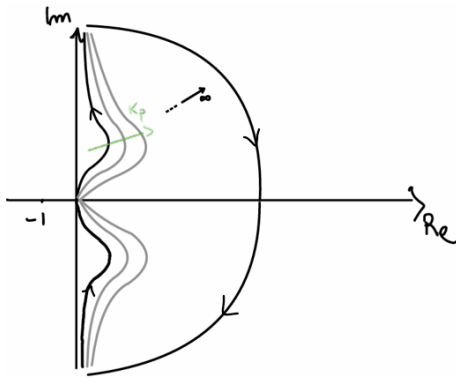
$$z < p_2$$



$$z > p_2$$



$$\begin{cases} \text{Phase margin } P_m > 0 \\ \text{Gain margin } G_m = +\infty \end{cases}$$



The stability is guaranteed for every value of k_p , the controlled system is ALWAYS stable.

Performances:

We study the performance of the controlled system by applying a unitary increase in speed with respect to its steady state condition ($V_{Ref} = 1 \text{ m/s}$), then by varying the value of k_p we look for an optimal configuration.

Performances are settled by studying some time domain's parameters. The search for performance is also based on the need to guarantee a system that provides acceptable comfort condition for the driver.

It also includes the choice of not having any type of overshoot, this is because we do not want the engine to find itself forced to apply a negative torque for having exceeded the reference.

The tolerances adopted are:

- Settling Time: $\leq 5 \text{ s}$ ($\pm 5\%$)
- Peak Overshoot: $\leq 5\%$
- Rise Time: 1 - 3 s
- Steady-State Error: $\leq 1 \text{ km/h}$ (3.6 m/s)

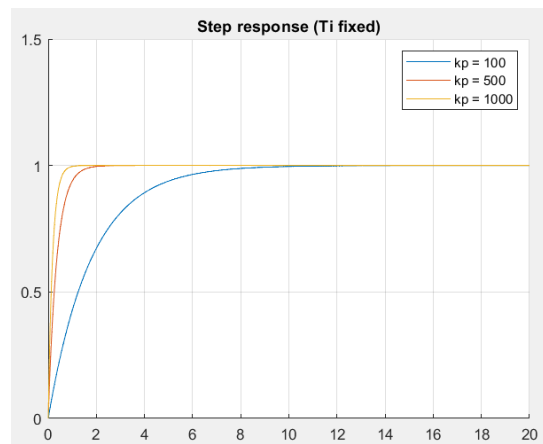
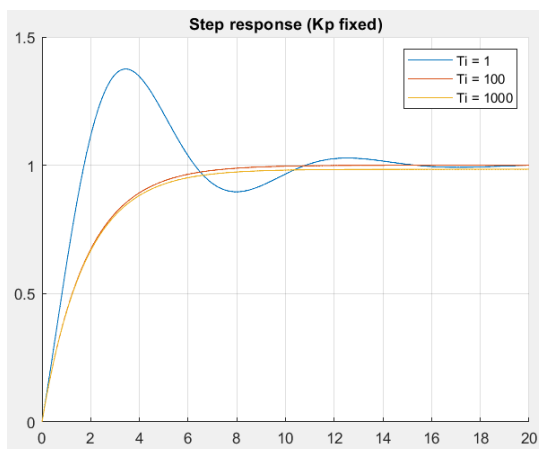
$$\text{For: } \begin{cases} k_p = 400 \\ T_i = 103 \end{cases}$$

Performance parameters:

$$\text{Time domain: } \begin{cases} \text{Settling Time} = 1.34 \\ \text{Rise Time} = 1.04 \\ \text{Steady - State Error} = 3.6927e - 05 \\ \text{Peak Overshoot} = 0 \end{cases}$$

Comments:

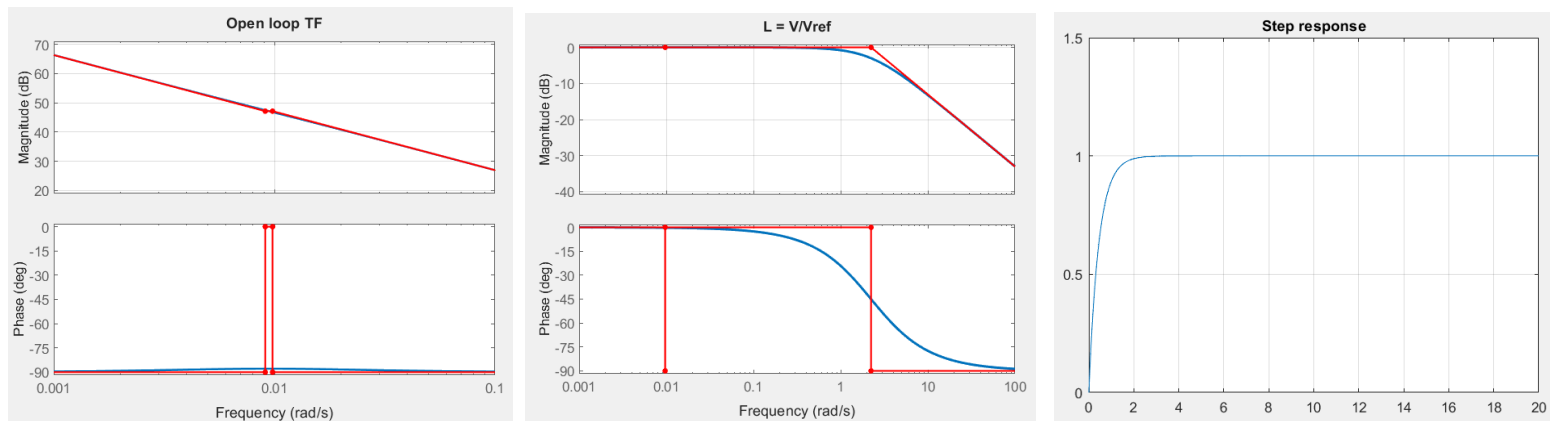
By some iterative analysis we observed that for $z > p_2$ (low T_i) undesired overshoot values may arise, that's why we preferred to place the zero before the pole.



Even though we had to keep the zero to the left of the pole, we decided to place it just before, in order to be able to exploit the contribution of the integral gain as much as possible.

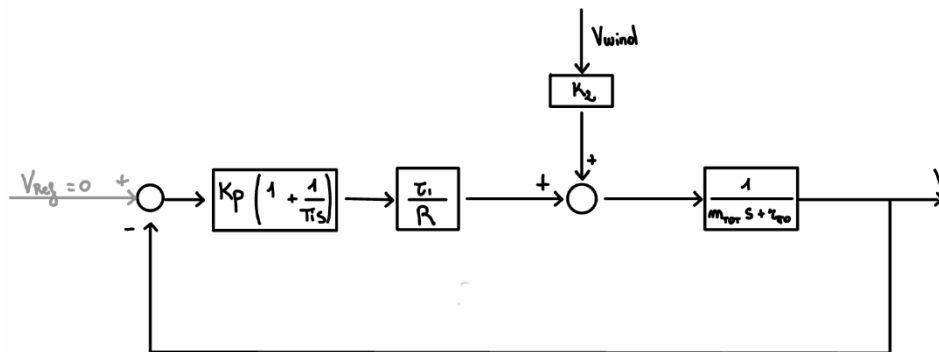
$$\left| \frac{1}{T_i} \right| < |p_2| = 0.0098 \Rightarrow T_i > \frac{1}{|p_2|} \cong 102$$

After that, we chose an optimal value of K_p in order to guarantee a reasonable value of the rise time and a bandwidth sufficiently high.



Time response due to external wind perturbation:

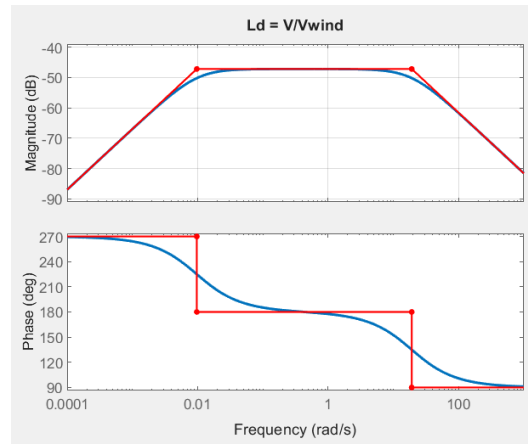
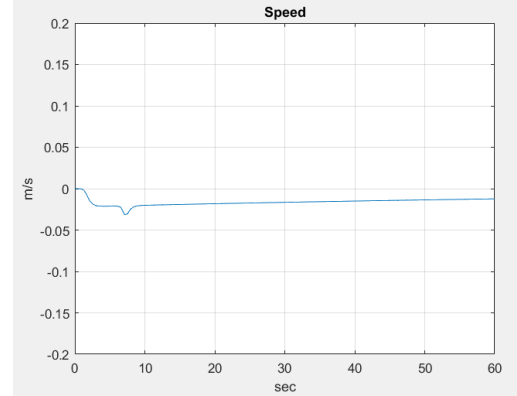
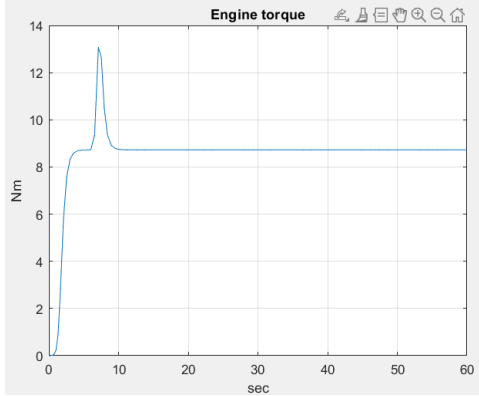
$$\frac{V}{V_{wind}} :$$



- Open loop: $K_2 K R \tilde{G}$
- Closed loop: $L_D(s) = \frac{V}{V_{wind}} = \frac{K_2 \tilde{G}}{1 + K R \tilde{G}}$

$$Speed_{wind}(t) = \frac{5}{1 + e^{-5(t-1.5)}} + 4e^{-\frac{(t-7)^2}{2 \cdot 0.3^2}}$$

The wind profile shown refers to the time variation with respect to the steady state value of the wind v_{wind_0} .



Part 3:

The control torque T_M is now provided by a permanent magnet DC motor:

$$\begin{cases} m^* \dot{v} + r^* v = T_M \frac{\tau_1}{R} - F_{Drag} - Mg \sin a - F_{Roll} \\ T_M = k_\phi i \\ V_a - k_\phi \omega_M = L_a \frac{di}{dt} + R_a i \end{cases} \quad \text{with } \omega_M = \frac{\tau_1 v}{R} \text{ angular speed of the motor}$$

V_{a_0} and i_0 values at steady state condition:

By imposing the steady state condition, we are able to find voltage and current's steady state values which are applied to the DC motor in order to provide a torque equal to T_{M_0} (whose value is already evaluated).

$$\begin{cases} v_{wind} = v_{wind_0} \\ v = v_0 \\ \dot{v} = 0 \\ T_c = 0 \\ i = i_0 \left(\frac{di}{dt} = 0 \right) \\ V_a = V_{a_0} \end{cases} \Rightarrow \begin{cases} r^* v_0 = T_{M_0} \frac{\tau_1}{R} - \frac{1}{2} \rho_A A_f C_x (v_0 + v_{wind_0})^2 - Mg \sin a - C_{RR} Mg \cos a (1 + k_{RR} v_0) \\ T_{M_0} = k_\phi i_0 \\ V_{a_0} - k_\phi \frac{\tau_1 v_0}{R} = R_a i_0 \end{cases}$$

$$V_{a_0} = \frac{R_a T_{M_0}}{k_\phi} + k_\phi \tau_1 \frac{v_0}{R} \Rightarrow V_{a_0} = 114.3 V$$

$$i_0 = \frac{T_{M_0}}{k_\phi} \Rightarrow i_0 = 228.8 A$$

Linearization in the neighbourhood of the steady state condition:

$$v = \tilde{v} + v_0 \quad \tilde{v} = \dot{v}$$

$$v_{wind} = v_{wind_0} - \tilde{v}_{wind} \quad (v_{wind} < 0)$$

$$i = i_0 + \tilde{i} \quad V_a = V_{a_0} + \tilde{V}_a$$

$F_{Drag} \cong F_{D_0} - K_1 \tilde{v} - K_2 \tilde{v}_{wind}$ while F_{Roll} is already a linear function.

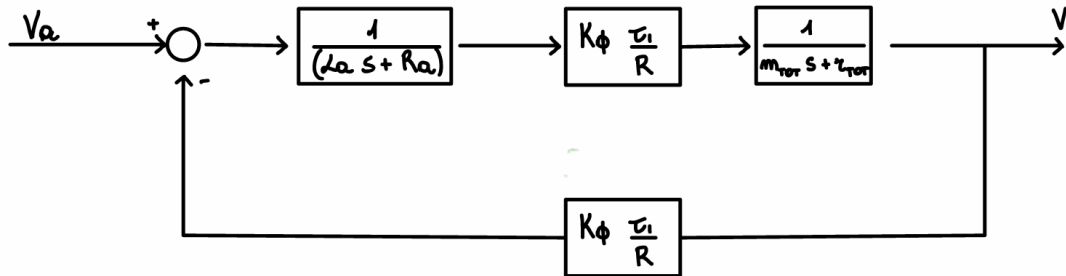
Linearized equation of motion of the electro-mechanical system:

$$\begin{cases} m_{TOT} \tilde{v} + r_{TOT} \tilde{v} = T_c \frac{\tau_1}{R} + K_2 \tilde{v}_{wind} \\ T_c = k_\phi \tilde{i} \\ \tilde{V}_a - k_\phi \frac{\tau_1 \tilde{v}}{R} = L_a \frac{d\tilde{i}}{dt} + R_a \tilde{i} \end{cases}$$

Laplace domain:

$$\Rightarrow \begin{cases} (m_{TOT} s + r_{TOT}) V = T_c \frac{\tau_1}{R} + K_2 V_{wind} \\ T_c = k_\phi I \\ V_a - k_\phi \frac{\tau_1 V}{R} = (L_a s + R_a) I \end{cases}$$

TF electro-mechanical system:



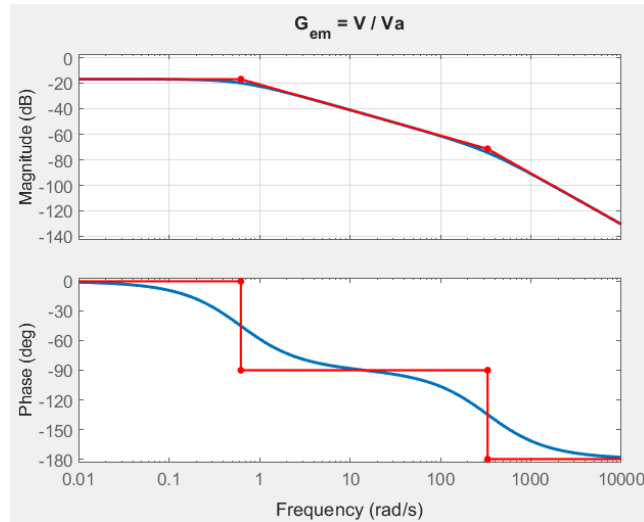
$$G_{em}(s) = \frac{V}{V_a} = \frac{k_\phi \frac{\tau_1}{R}}{(m_{TOT} s + r_{TOT})(L_a s + R_a) + \left(k_\phi \frac{\tau_1}{R}\right)^2}$$

$$= \frac{k_\phi \frac{\tau_1}{R}}{(m_{TOT} L_a) s^2 + (m_{TOT} R_a + r_{TOT} L_a) s + (r_{TOT} R_a + \left(k_\phi \frac{\tau_1}{R}\right)^2)}$$

2 poles:

$$p_{1,2} = \frac{-(m_{TOT} R_a + r_{TOT} L_a) \pm \sqrt{(m_{TOT} R_a + r_{TOT} L_a)^2 - 4(m_{TOT} L_a)(r_{TOT} R_a + \left(k_\phi \frac{\tau_1}{R}\right)^2)}}{2(m_{TOT} L_a)} = \begin{cases} p_1 = -0.6231 \\ p_2 = -332.7201 \end{cases}$$

We have two stable poles, one comes from the mechanical system contribution, the other from the DC motor contribution; the uncontrolled electro-mechanical system is stable



$T_c(t)$ is now governed by a PID regulator acting on the armature voltage $V_a(t)$:

$$V_a(t) = k_p(v_{Ref} - \tilde{v}) + k_d(\dot{v}_{Ref} - \dot{\tilde{v}}) + k_i \left(\int_0^{+\infty} (v_{Ref} - \tilde{v}) dt \right)$$

Laplace domain:

$$V_a = k_p(V_{Ref} - V) + k_d s(V_{Ref} - V) + \frac{k_i}{s}(V_{Ref} - V)$$

$$k_d = k_p T_d \quad k_i = \frac{k_p}{T_i} \quad V_{Ref} = 0$$

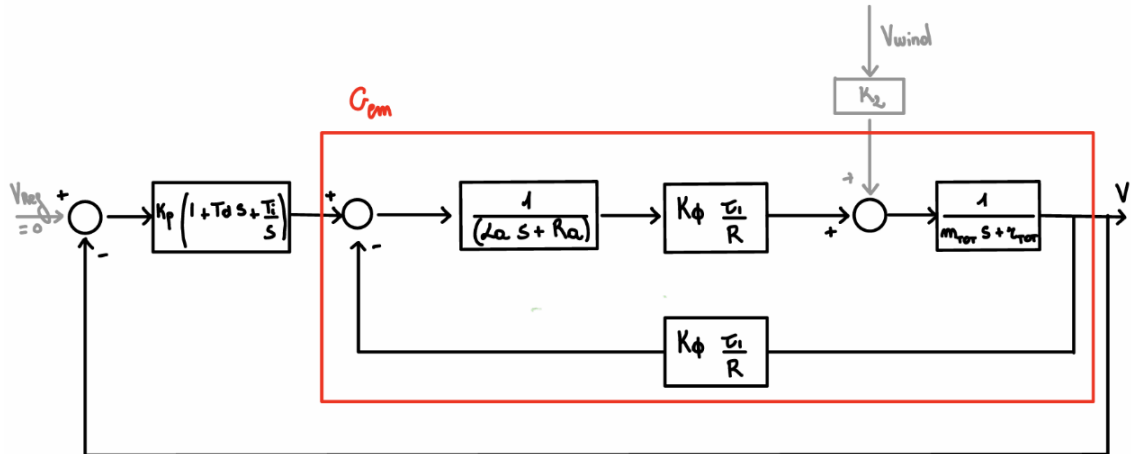
$$\Rightarrow \begin{cases} (m_T s + r_T)V = T_C \frac{\tau_1}{R} + K_2 V_{wind} \\ T_c = k_\phi I \\ V_a - k_\phi \frac{\tau_1 V}{R} = (L_a s + R_a)I \\ V_a = \left(k_p + k_d s + \frac{k_i}{s} \right) (-V) = -k_p \left(1 + T_d s + \frac{1}{T_i s} \right) V \end{cases}$$

Stability analysis:

We define:

$$k_p \left(1 + T_d s + \frac{1}{T_i s} \right) = R$$

$$\frac{V}{V_{Ref}} :$$



- Open loop: $G_{em}R = \frac{[(T_d T_i) s^2 + T_i s + 1] k_p k_\phi \frac{\tau_1}{R}}{[(m_{TOT} L_a) s^3 + (m_{TOT} R_a + r_{TOT} L_a) s^2 + (r_{TOT} R_a + (k_\phi \frac{\tau_1}{R})^2) s] T_i}$

$$2 \text{ zeros: } z_{1,2} = \frac{-T_i \pm \sqrt{T_i^2 - 4T_i T_d}}{2T_i T_d} = \frac{-1}{2T_d} \pm \sqrt{\frac{1}{(2T_d)^2} - \frac{1}{T_d T_i}}$$

The zeros can be:

- Complex conjugate if $T_i^2 - 4T_i T_d < 0 \Rightarrow 4T_d > T_i$
- Both real and equal to the same value if $T_i^2 - 4T_i T_d = 0 \Rightarrow 4T_d = T_i$
- Both real but with different values if $T_i^2 - 4T_i T_d > 0 \Rightarrow 4T_d < T_i$

3 poles: $p_1 = 0$, $p_2 = -0.6231$, $p_3 = -332.7201$ (stable)

- Closed loop: $L(s) = \frac{V}{V_{Ref}} = \frac{G_{em}R}{1 + G_{em}R}$

$$= \frac{[(T_d T_i) s^2 + T_i s + 1] k_p k_\phi \frac{\tau_1}{R}}{(m_{TOT} L_a T_i) s^3 + [(m_{TOT} R_a + r_{TOT} L_a) T_i + T_d T_i k_p k_\phi \frac{\tau_1}{R}] s^2 + \left[(r_{TOT} R_a + (k_\phi \frac{\tau_1}{R})^2) T_i + T_i k_p k_\phi \frac{\tau_1}{R} \right] s + k_p k_\phi \frac{\tau_1}{R}}$$

2 zeros and 3 poles

We analyse the stability of the system by applying some undirect and direct methods (margin, Nyquist and root locus)

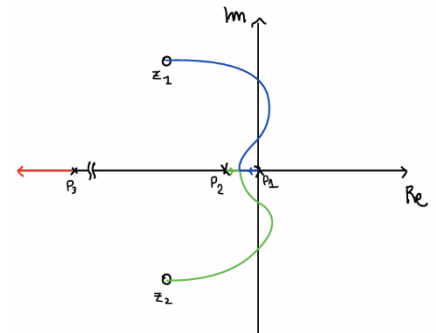
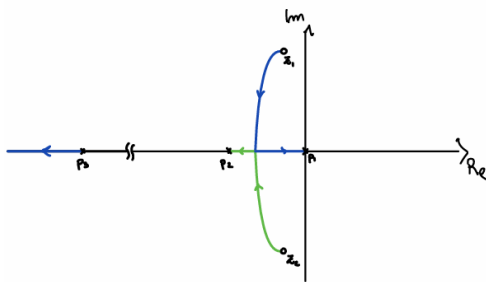
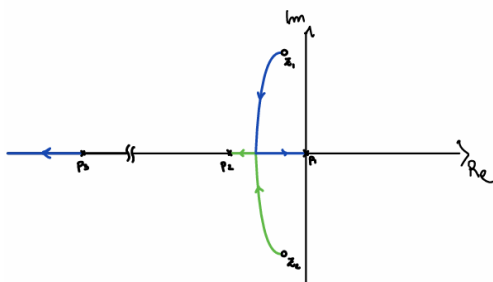
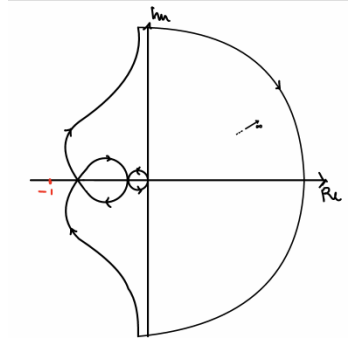
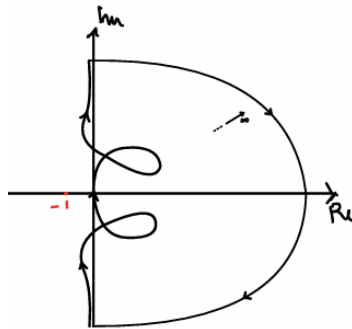
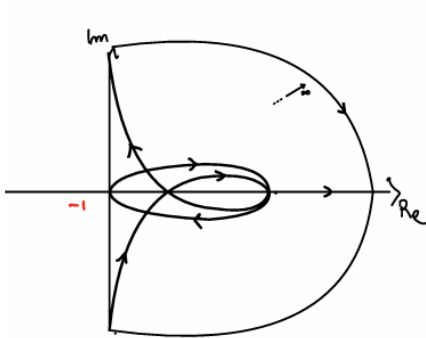
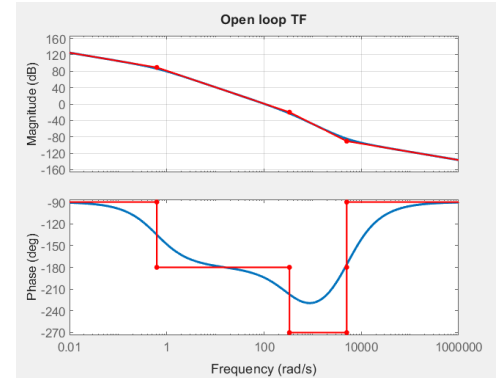
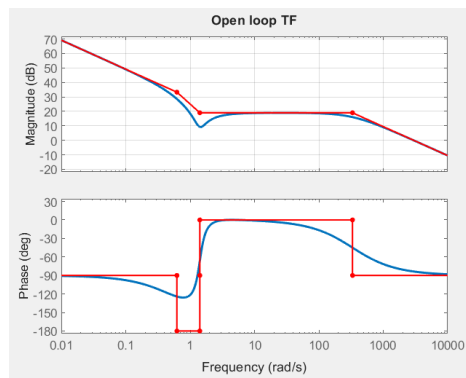
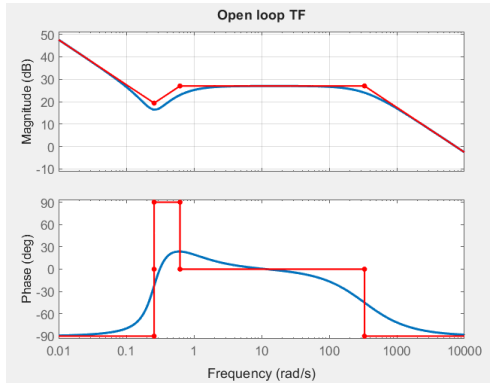
Here are some possible conditions that may arise depending on the position of the two zeros:

$T_i < 4T_d$ (complex conjugate zeros):

$$z_{1,2} < p_2$$

$$p_2 < z_{1,2} < p_3$$

$$z_{1,2} > p_3$$

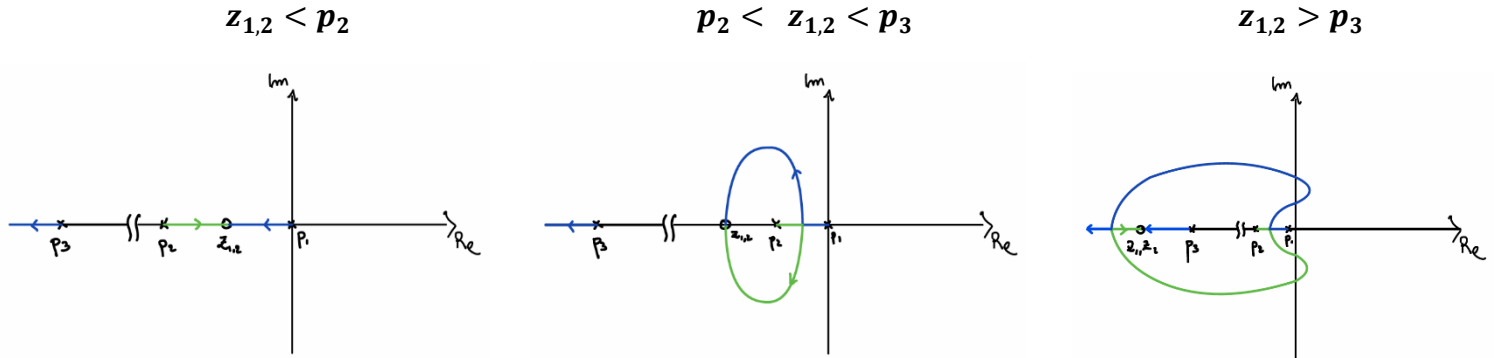


For $z_{1,2} < p_2$ and $p_2 < z_{1,2} < p_3$ the controlled system is always stable for every k_p adopted, however there may be situations where for some k_p the poles are complex conjugate and so we'll have oscillations of the system (not likable).

However, if $z_{1,2} > p_2$ the system gets unstable for a limited range of k_p .

$$T_i = 4T_d \text{ (real equal zeros): } z_{1,2} = -\frac{1}{2T_d}$$

The Bode and Nyquist diagrams are basically the same as the previous case, only the root locus will change:

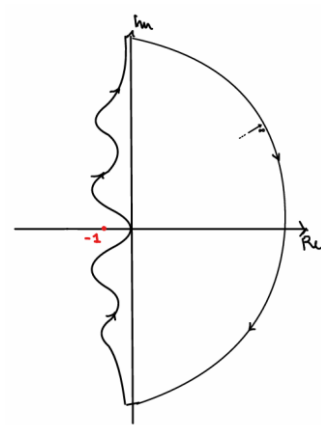
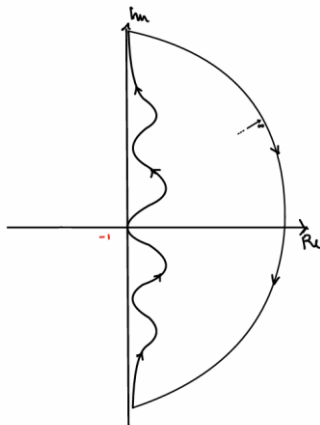
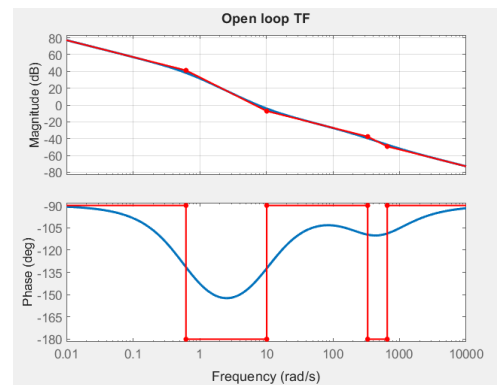
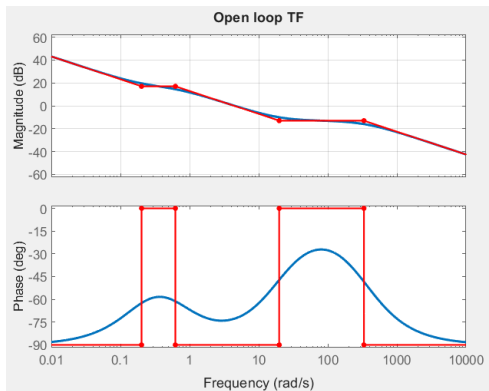


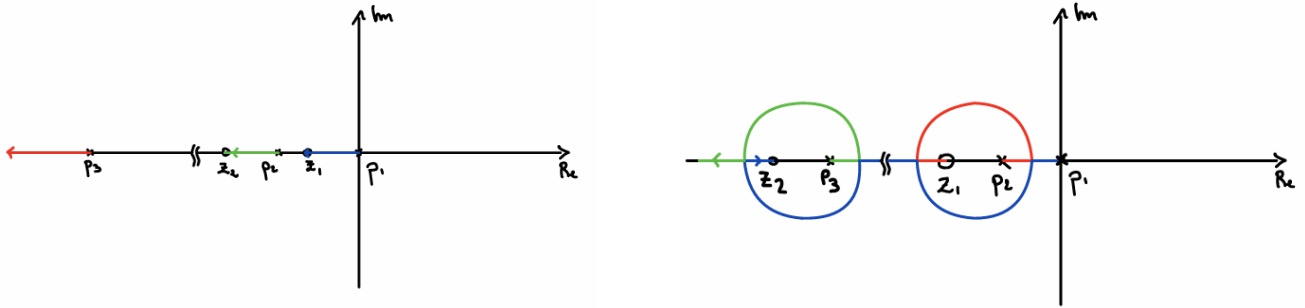
As for the previous case, if $z_{1,2} > p_3$ the controlled system can be unstable for a limited range of k_p values.

$$T_i > 4T_d \text{ (Real distinct zeros):}$$

$$z_1 < p_2, p_2 < z_2 < p_3$$

$$p_2 < z_1 < p_3, z_2 < p_3$$





The controlled system is always stable for every value of k_p

Performances:

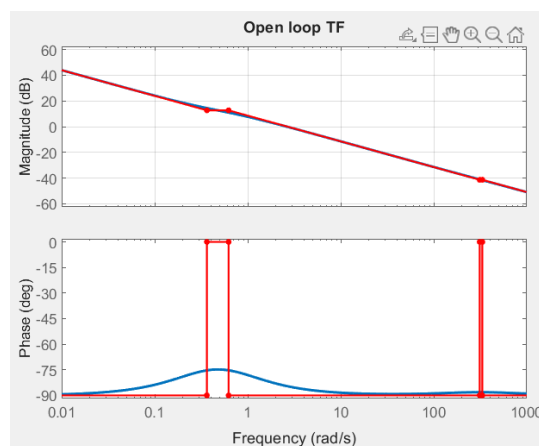
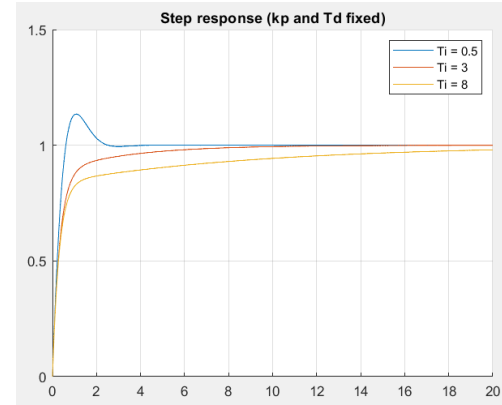
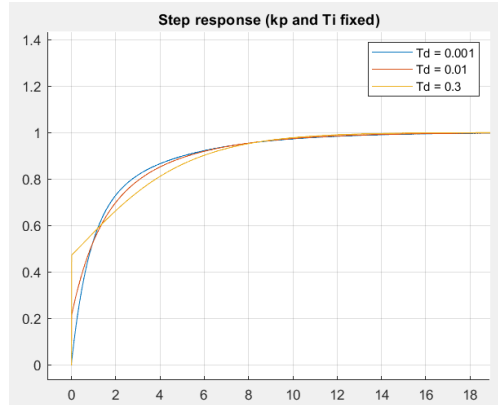
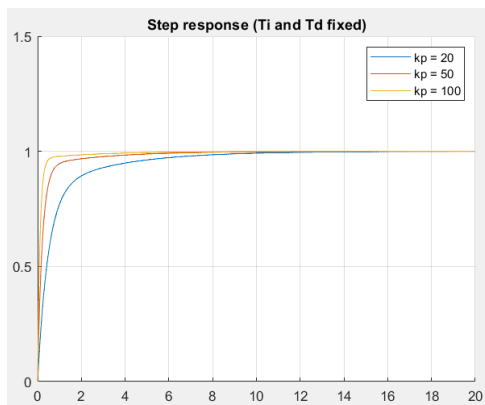
Performances are settled by studying the behaviour of the system in front of a speed variation of 1 m/s ($V_{Ref} = 1 \text{ m/s}$).

$$\text{For: } \begin{cases} k_p = 30 \\ T_i = 2.75 \\ T_d = 0.0032 \end{cases}$$

Performance parameters:

$$\text{Time domain: } \begin{cases} \text{Settling Time} = 2.47 \text{ sec} \\ \text{Rise Time} = 1.21 \text{ sec} \\ \text{Steady - State Error} = 0.00015806 \\ \text{Peak Overshoot} = 0 \end{cases}$$

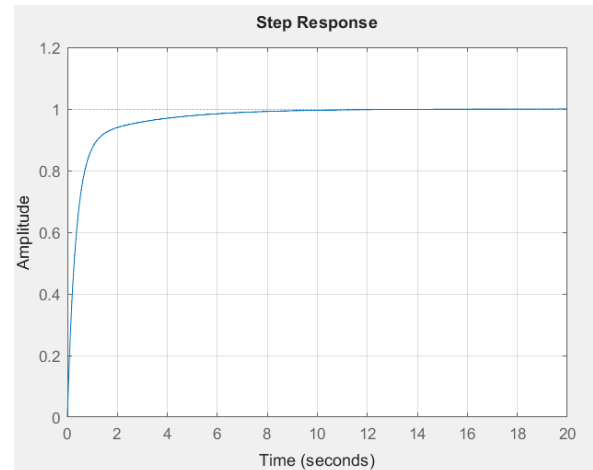
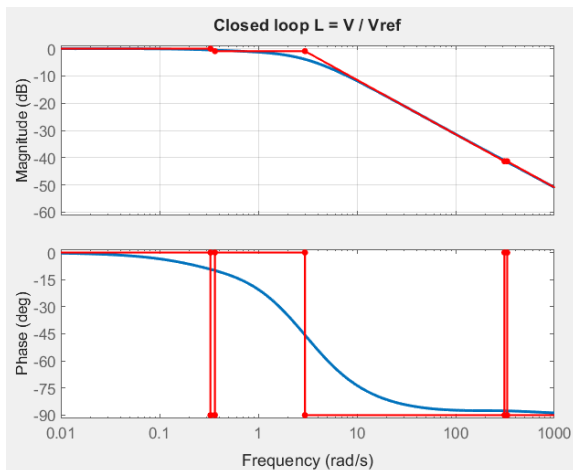
Comments:



We decided to choose a combination of T_i and T_d in order to have two real zeros with distinct values ($T_i > 4T_d$); then each zero was placed just before each of p_2 and p_3

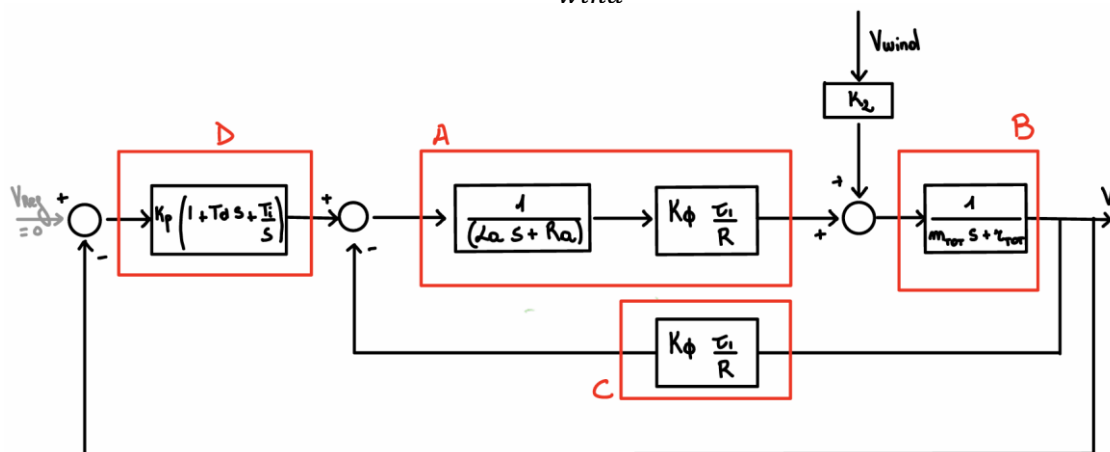
This condition is also one of the few that allows the poles of L to be always real for any value of k_p .

Although the performances were studied on a step function (whose increase is almost instantaneous and therefore unrealistic), it was decided to maintain a low level of k_d (low T_d) so as not to have a rise time excessively low.

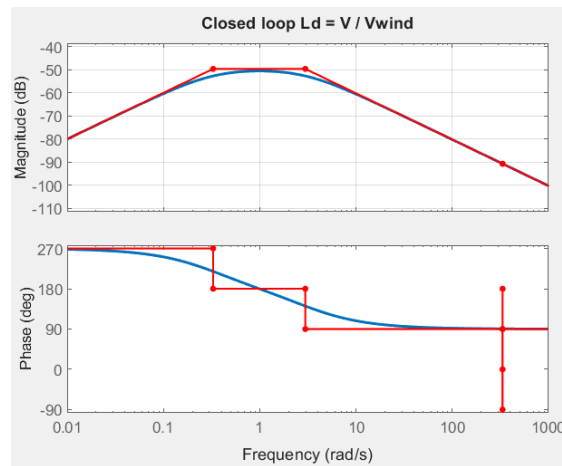
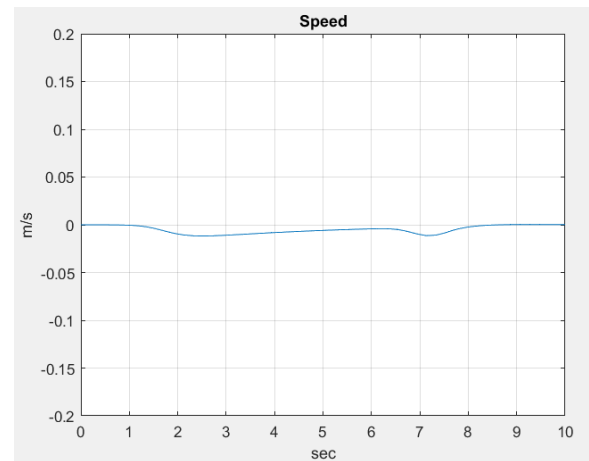
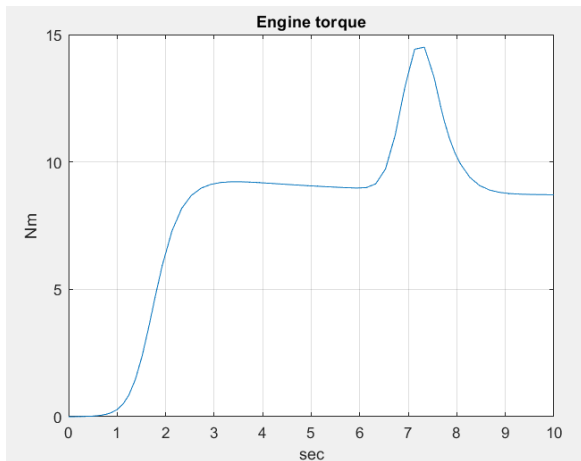


Time response due to external wind perturbation:

$$\frac{V}{V_{wind}} :$$



- Closed loop: $L_D(s) = \frac{V}{V_{wind}} = \frac{\left(\frac{+K_2B}{1+BAC}\right)}{\left(1+\frac{B}{1+BAC}AD\right)}$

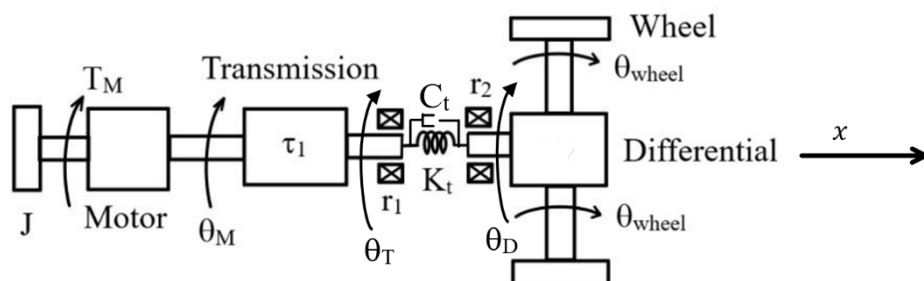


Part 4

The drive shaft is now deformable: $k_t = \frac{400Nm}{rad}$, $c_t = \frac{4Nms}{rad}$

The system has 2 dof.:

- The displacement of the vehicle x
- The angular displacement of the motor θ_M



Lagrange approach multi-dof system (matrix notation):

We define $\underline{z} = \begin{bmatrix} \theta_M \\ x \end{bmatrix}$

$$E_k = \frac{1}{2} \dot{\underline{z}}^T \begin{bmatrix} J_m & 0 \\ 0 & \frac{J_\omega}{R^2} 4 + M \end{bmatrix} \dot{\underline{z}} = \frac{1}{2} \dot{\underline{z}}^T [M] \dot{\underline{z}} \quad V = \frac{1}{2} \underline{z}^T \begin{bmatrix} \frac{k_t}{\tau_1^2} & -\frac{k_t}{\tau_1 R} \\ -\frac{k_t}{\tau_1 R} & \frac{k_t}{R^2} \end{bmatrix} \underline{z} = \frac{1}{2} \underline{z}^T [K] \underline{z}$$

$$D = \frac{1}{2} \dot{\underline{z}}^T \begin{bmatrix} \frac{r_1 + c_t}{\tau_1^2} & -\frac{c_t}{\tau_1 R} \\ -\frac{c_t}{\tau_1 R} & \frac{r_2 + c_t}{R^2} \end{bmatrix} \dot{\underline{z}} = \frac{1}{2} \dot{\underline{z}}^T [C] \dot{\underline{z}} \quad \delta \underline{W} = \begin{bmatrix} -F_{Drag} - Mg \sin a - F_{Roll} \end{bmatrix} \delta \underline{z} = \underline{Q}^T \delta \underline{z}$$

We get the expression for the non-linear equation of motion of the system

$$[M] \ddot{\underline{z}} + [C] \dot{\underline{z}} + [K] \underline{z} = \underline{Q}$$

Linearization in the neighbourhood of the steady state condition:

$$\dot{\theta}_M = \dot{\theta}_{M_0}, \dot{x} = v_0, v_{wind} = v_{wind_0}$$

$$x = x_{ss} + \tilde{x} \quad \dot{x} = v_0 + \tilde{\dot{x}} \quad \ddot{x} = \tilde{\ddot{x}}$$

$$\theta_M = \theta_{M_{ss}} + \tilde{\theta}_M \quad \dot{\theta}_M = \dot{\theta}_{M_0} + \tilde{\dot{\theta}}_M \quad \ddot{\theta}_M = \tilde{\ddot{\theta}}_M$$

$$v_{wind} = v_{wind_0} - \tilde{v}_{wind} \quad (v_{wind} < 0)$$

$$F_{Drag} \cong F_{D_0} - K_1 \tilde{v} - K_2 \tilde{v}_{wind} \quad \text{while } F_{Roll} \text{ is already a linear function.}$$

Linearized equation of motion of the system:

$$\begin{bmatrix} J_m & 0 \\ 0 & \frac{J_\omega}{R^2} 4 + M \end{bmatrix} \ddot{\underline{\tilde{z}}} + \begin{bmatrix} \frac{r_1 + c_t}{\tau_1^2} & -\frac{c_t}{\tau_1 R} \\ -\frac{c_t}{\tau_1 R} & \left(\frac{r_2 + c_t}{R^2} - K_1 + C_{RR} Mg \cos a k_{RR} \right) \end{bmatrix} \dot{\underline{\tilde{z}}} + \begin{bmatrix} \frac{k_t}{\tau_1^2} & -\frac{k_t}{\tau_1 R} \\ -\frac{k_t}{\tau_1 R} & \frac{k_t}{R^2} \end{bmatrix} \underline{\tilde{z}} = \begin{bmatrix} T_c \\ +K_2 \tilde{v}_{wind} \end{bmatrix}$$

$$[M] \ddot{\underline{\tilde{z}}} + [C] \dot{\underline{\tilde{z}}} + [K] \underline{\tilde{z}} = \underline{f}$$

Laplace domain:

$$\Rightarrow ([M]s^2 + [C]s + [K])\underline{Z} = \underline{F}$$

We define $([M]s^2 + [C]s + [K]) = [E]$; the transfer functions of the 2 dof passive system are collected in $[\tilde{G}] = [E]^{-1} = \frac{1}{\det[E]} \begin{bmatrix} E_{22} & -E_{12} \\ -E_{21} & E_{11} \end{bmatrix}$

$\det[E]$ is a 4th order polynomial

$$\det[E] = s[as^3 + bs^2 + cs + d]$$

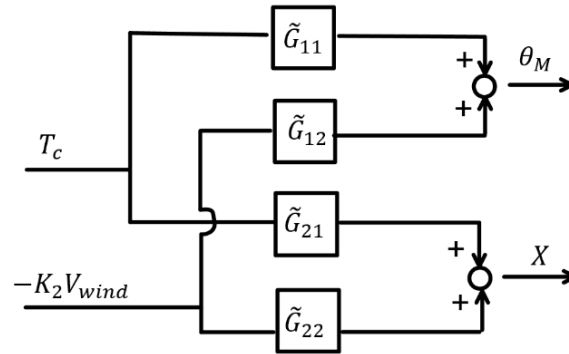
$$= s[M_{11}M_{22}s^3 + (M_{11}R_{22} + M_{22}R_{11} - R_{21}^2)s^2 + (M_{11}K_{22} + M_{22}K_{11} + R_{11}R_{22} - 2R_{21}K_{21})s + (R_{11}K_{22} + R_{22}K_{11})]$$

The roots of $\det[E]$ are basically the poles of the uncontrolled system's transfer functions.

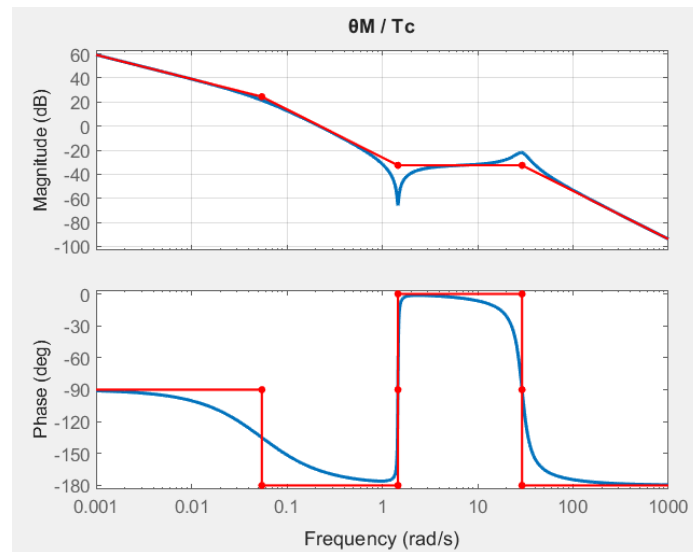
$$p_{1,2,3,4} = \begin{cases} 0.0000 + 0.0000i \\ -4.3435 + 28.8804i \\ -4.3435 - 28.8804i \\ -0.0547 + 0.0000i \end{cases}$$

We notice that $p_1 = 0$ as the system is labile, however, all the real parts of the poles are negative, so the uncontrolled system is stable

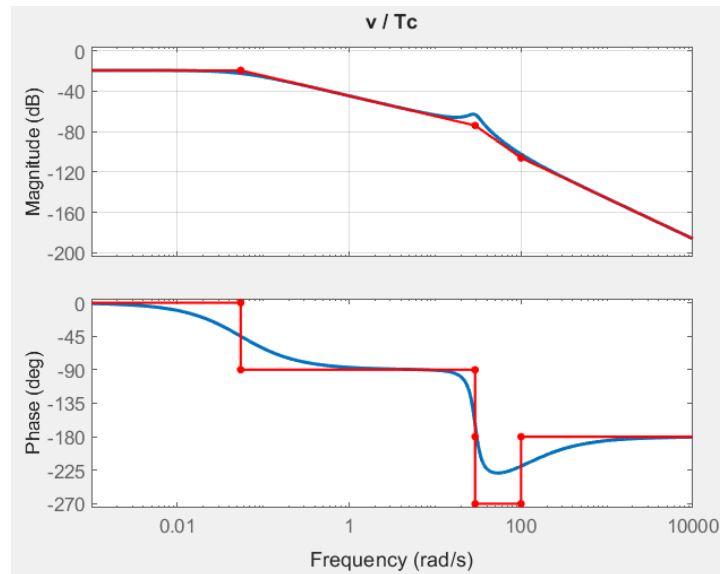
Transfer functions:



$$\tilde{G}_{11}(s) = \frac{\theta_M(s)}{T_c(s)} = \frac{E_{22}}{\det[E]} = \frac{\left(\frac{J_\omega}{R^2} 4 + M\right) s^2 + \left(\frac{r_2 + c_t}{R^2} + K_1 + C_{RR} M g \cos a k_{RR}\right) s + \frac{k_t}{R^2}}{\det[E]}$$



$$\tilde{G}_{21}(s) = \frac{x(s)}{T_c(s)} \rightarrow \frac{v(s)}{T_c(s)} = \tilde{G}_{21}s = \frac{\left(\frac{c_t}{\tau_1 R}s + \frac{k_t}{\tau_1 R}\right)}{\det[E]}s$$



$T_C(t)$ is now governed by a PID regulator; we'd like to control the speed of the vehicle, however, this time the system has two separate dofs.

$$T_C(t) = k_p(v_{Ref} - \tilde{v}) + k_d(\dot{v}_{Ref} - \dot{\tilde{v}}) + k_i \left(\int_0^{+\infty} (v_{Ref} - \tilde{v}) dt \right)$$

The torque isn't applied directly to the degree of freedom we want to control; we are dealing with a non co-located control.

Laplace domain:

$$T_C = k_p(V_{Ref} - V) + k_d s(V_{Ref} - V) + \frac{k_i}{s}(V_{Ref} - V)$$

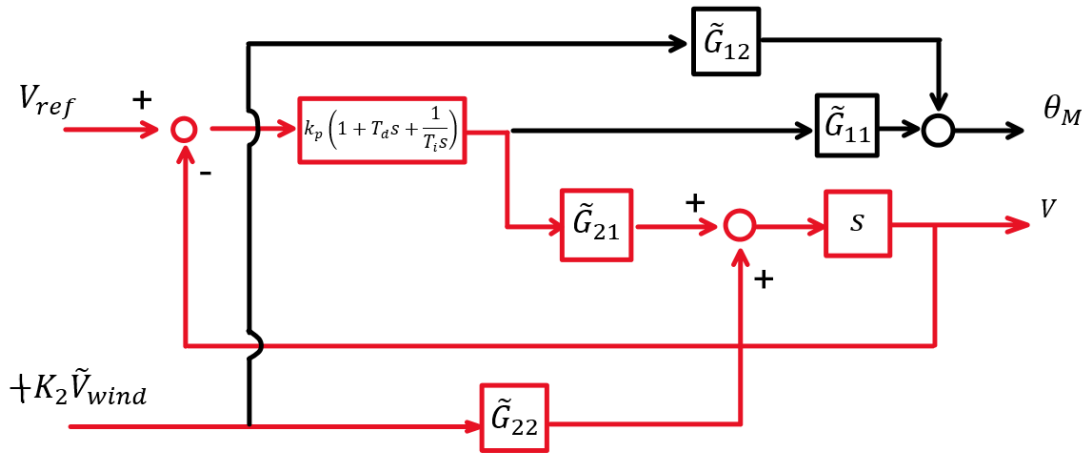
$$k_d = k_p T_d \quad k_i = \frac{k_p}{T_i} \quad V_{Ref} = 0$$

$$\begin{cases} ([M]s^2 + [C]s + [K])\underline{Z} = \begin{bmatrix} T_C \\ +K_2 V_{wind} \end{bmatrix} \\ T_C = -k_p \left(1 + T_d s + \frac{1}{T_i s} \right) V \end{cases}$$

Stability analysis:

We define:

$$k_p \left(1 + T_d s + \frac{1}{T_i s} \right) = R$$



$$\frac{V}{V_{Ref}} :$$

- *Open loop:* $R\tilde{G}_{21}s = k_p \left(1 + T_d s + \frac{1}{T_i s} \right) \frac{\left(\frac{c_t}{\tau_1 R} s + \frac{k_t}{\tau_1 R} \right)}{\det[E]} s$

$$3 \text{ zeros: } z_{1,2} = \frac{-T_i \pm \sqrt{T_i^2 - 4T_i T_d}}{2T_i T_d}, \quad z_3 = -\frac{k_t}{c_t} = -100$$

$z_{1,2}$ can be:

- Complex conjugate if $T_i^2 - 4T_i T_d < 0 \Rightarrow 4T_d > T_i$
- Both real and equal to the same value if $T_i^2 - 4T_i T_d = 0 \Rightarrow 4T_d = T_i$
- Both real but with different values if $T_i^2 - 4T_i T_d > 0 \Rightarrow 4T_d < T_i$

$$4 \text{ poles: } \begin{cases} p_1 = 0 \\ p_2 = -0.0547 \\ p_{3,4} = -4.3435 \pm 28.8804i \end{cases}$$

- *Closed loop:* $L(s) = \frac{V}{V_{Ref}} = \frac{R\tilde{G}_{21}s}{1 + R\tilde{G}_{21}s}$

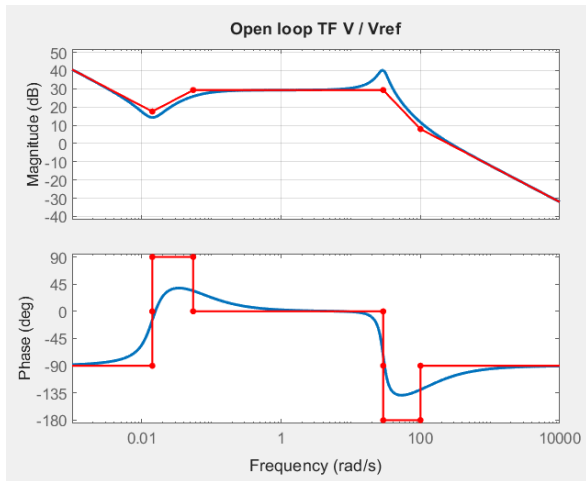
3 zeros and 4 poles

We analyse the stability of the system by applying some undirect and direct methods (margin, Nyquist and root locus)

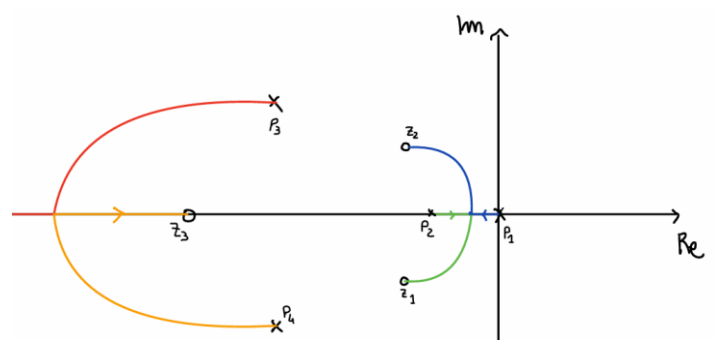
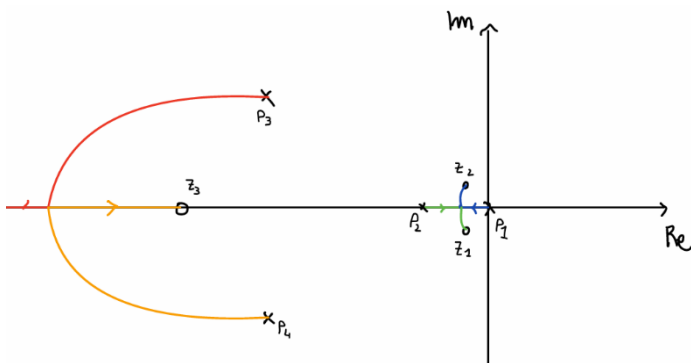
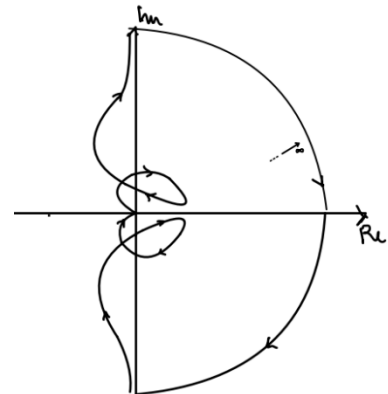
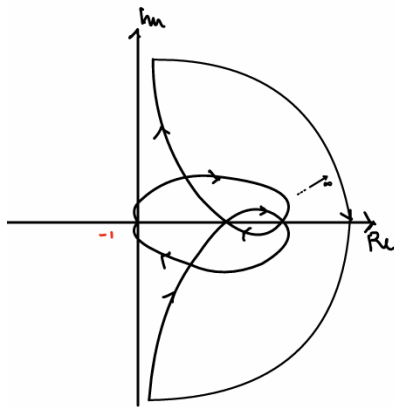
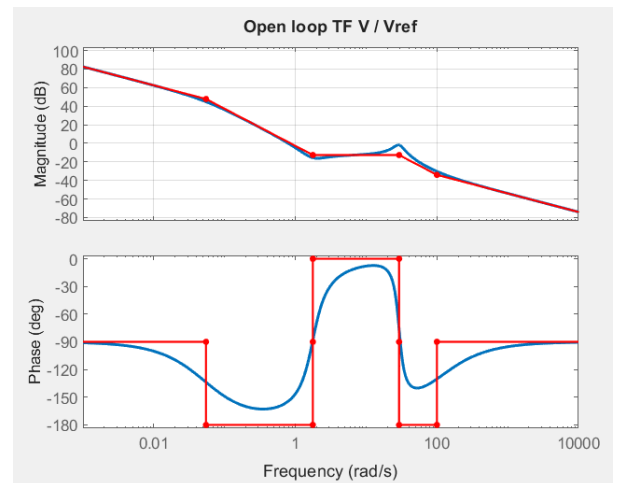
Here are some possible conditions that may arise depending on the position of the two zeros $z_{1,2}$:

$T_i < 4T_d$ (complex conjugate zeros):

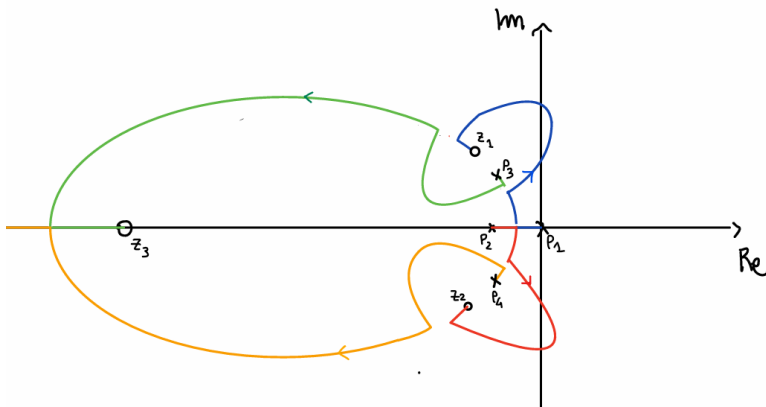
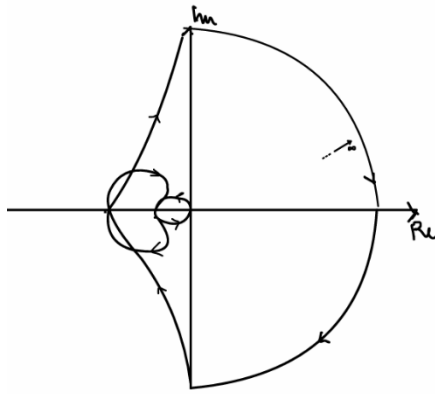
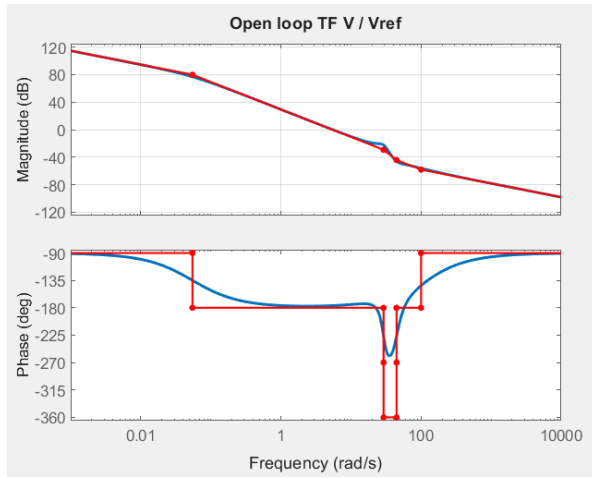
$$p_1 < z_{1,2} < p_2$$



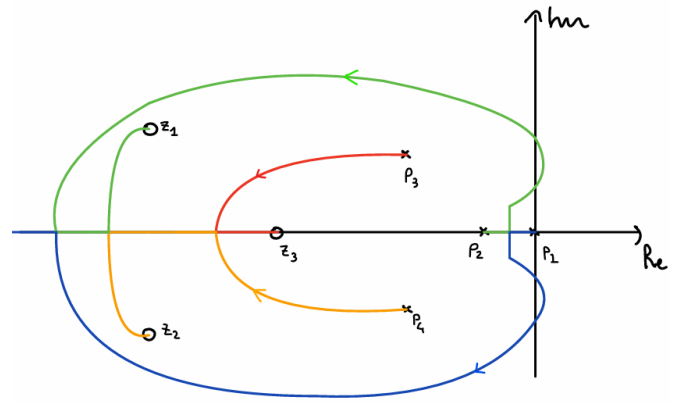
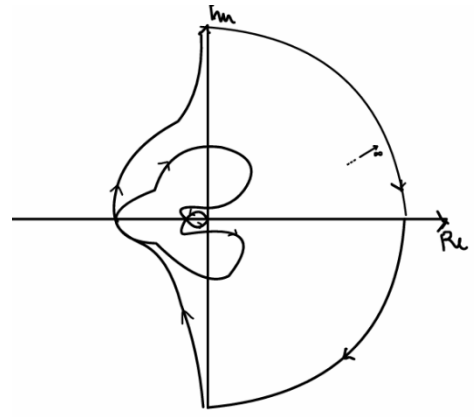
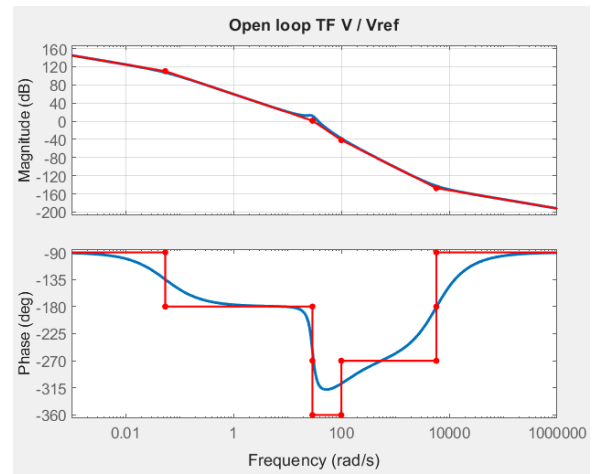
$$p_2 < z_{1,2} < p_{3,4}$$



$$p_{3,4} < z_{1,2} < z_3$$



$$z_{1,2} > z_3$$



For $z_{1,2} < p_{3,4}$ the controlled system is always stable for every k_p adopted, however there may be situations where for some k_p the poles are complex conjugate and so we'll have oscillations of the system (not likable).

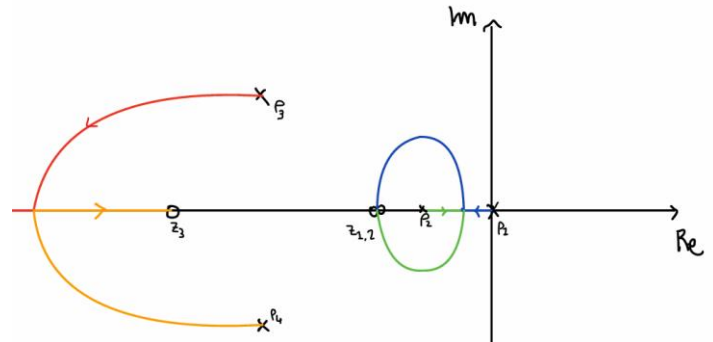
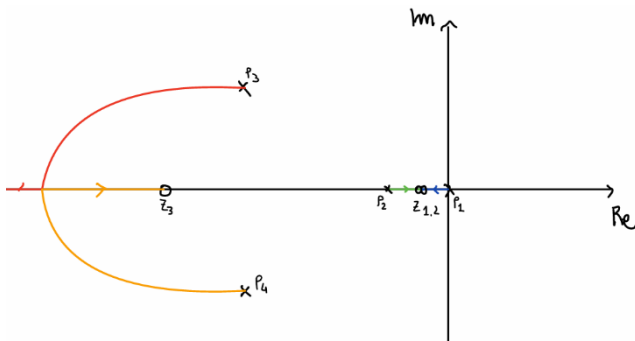
However, if $z_{1,2} > p_{3,4}$ the system gets unstable for a limited range of k_p .

$$T_i = 4T_d \text{ (real coincident zeros): } z_{1,2} = -\frac{1}{2T_d}$$

The Bode and Nyquist diagrams are basically the same as the previous cases, only the root locus will change:

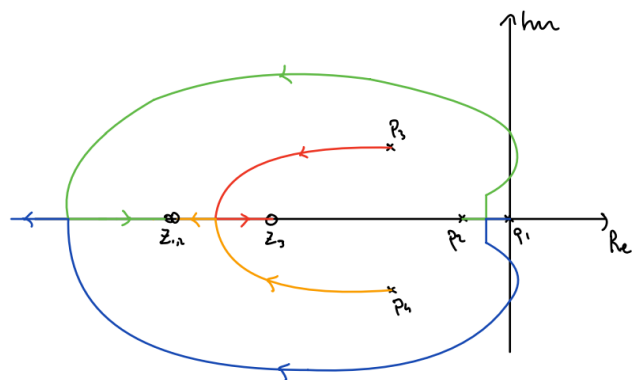
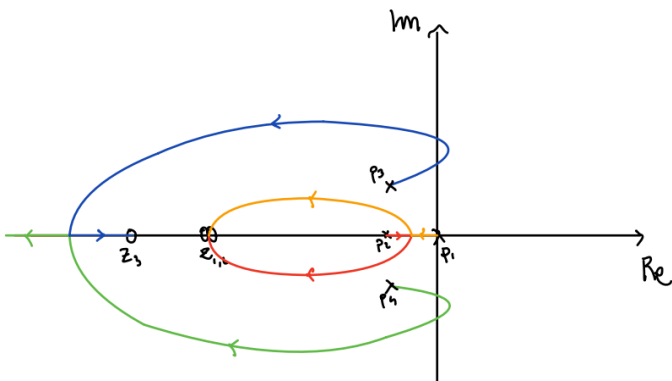
$$p_1 < z_{1,2} < p_2$$

$$p_2 < z_{1,2} < p_{3,4}$$



$$p_{3,4} < z_{1,2} < z_3$$

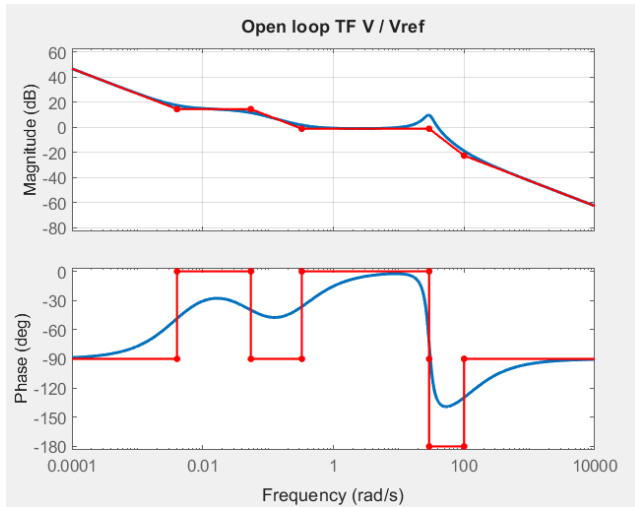
$$z_{1,2} > z_3$$



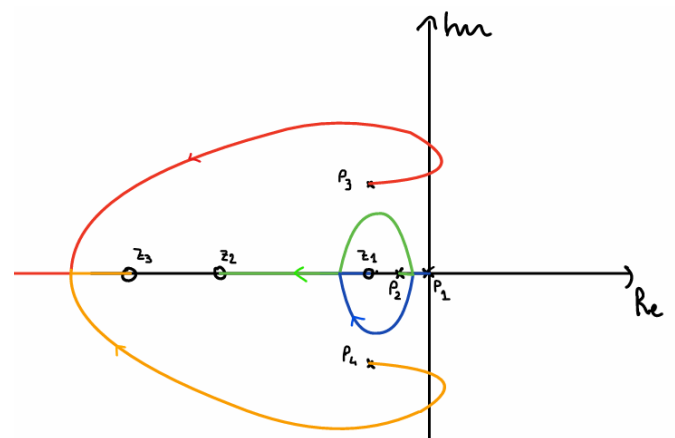
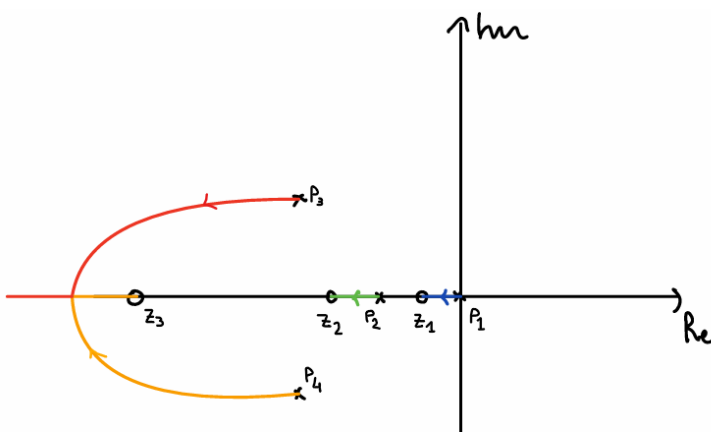
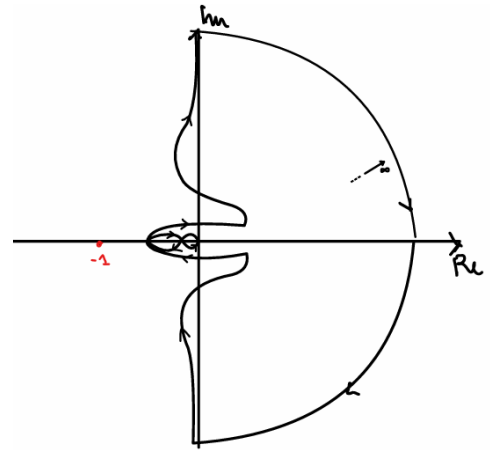
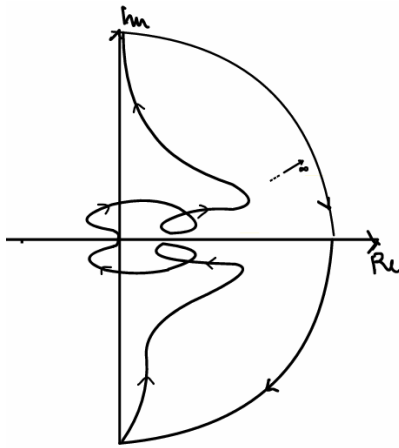
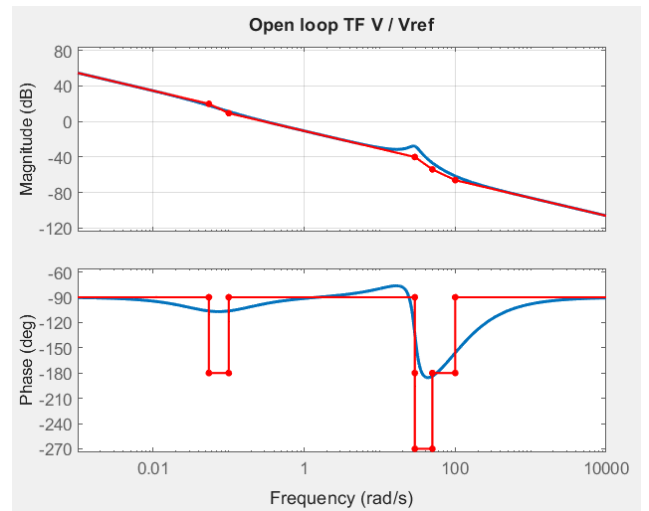
As for the previous case, if $z_{1,2} > p_{3,4}$ the controlled system can be unstable for a limited range of k_p values

$T_i > 4T_d$ (Real distinct zeros):

$$p_1 < z_1 < p_2, p_2 < z_2 < p_{3,4}$$



$$p_2 < z_1 < p_{3,4}, p_{3,4} < z_2 < z_3$$



The controlled system is always stable for the first case, however if we move the zeros towards the right and we consider the second case, there may be some unstable situations for a limited range of k_p .

Performances:

Performances are settled by studying the behaviour of the system in front of a speed variation of 1 m/s ($V_{Ref} = 1 \text{ m/s}$).

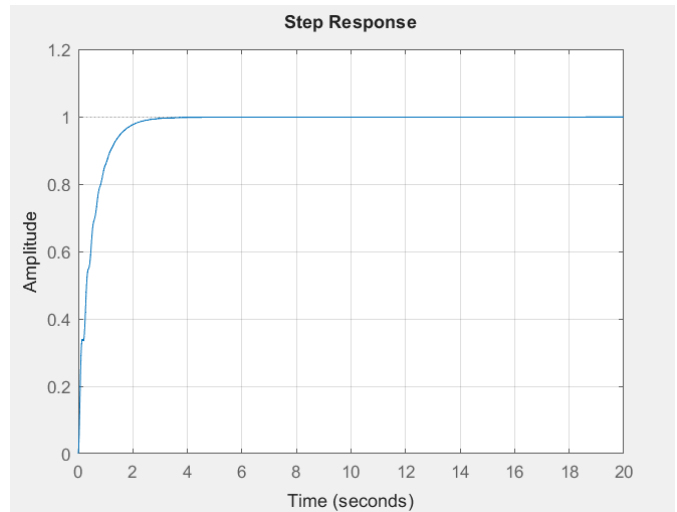
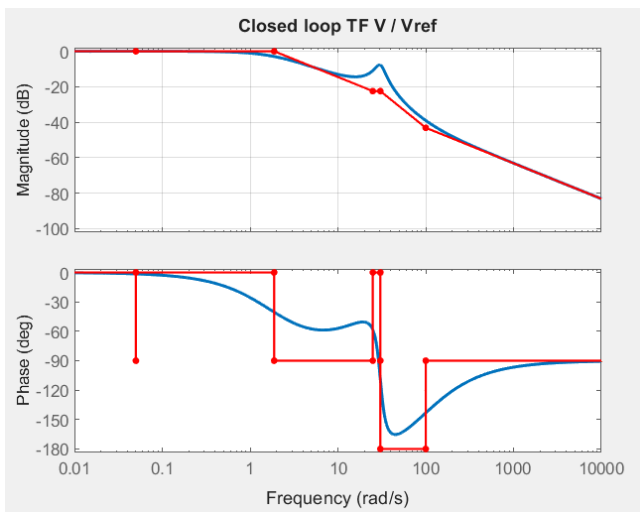
$$\text{For: } \begin{cases} k_p = 350 \\ T_i = 20 \\ T_d = 0.04 \end{cases}$$

Performance parameters:

$$\text{Time domain: } \begin{cases} \text{Settling Time} = 1.56 \text{ sec} \\ \text{Rise Time} = 1.19 \text{ sec} \\ \text{Steady - State Error} = 0.0008643 \\ \text{Peak Overshoot} = 0 \end{cases}$$

The first goal is to make sure that the controlled system has all its poles real and negative, however this is a condition which is hard to achieve because in order to make the poles $p_{3,4}$ real a very large value of k_p is needed independently from the position of the zeros $z_{1,2}$, causing the system to be extremely fast with respect to the tolerances adopted.

We decided to place the two zeros in order to have $p_1 < z_1 < p_2$, $p_2 < z_2 < p_{3,4}$ (the zeros are both real and with distinct values, $T_i > 4T_d$).



The step response shows some initial oscillations, as the poles $p_{3,4}$ of the closed loop TF are complex conjugate

Time response due to wind perturbation:

We are now studying the behaviour of the system in front of a unitary wind speed gain. The values of k_p, T_i, T_d are the same as the one used for the reference performances.

($V_{wind} = 1 \text{ m/s}$):

The transfer function we need to look at is:

$$\frac{V}{V_{wind}} :$$

- Closed loop: $Ld(s) = \frac{V}{V_{wind}} = \frac{K_2 \tilde{G}_{22}s}{1 + R\tilde{G}_{22}s}$

